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EE499

Research Project Final Report

Wearable Surface EMG for Detecting Bicep Contraction

1. Literature Review

Muscle fatigue is a very important factor in exercise and it basically dictates how an athlete performs and recovers. When a muscle becomes fatigued, it leads to the muscle losing strength and control which can lead to poor form or possibly injury. From research I have found, most of the regular methods for measuring muscle fatigue are done in labs using wired electromyography (EMG) systems and motion capture equipment. While these systems are probably the most accurate form, they are expensive and not very practical for everyday use. There is growing interest in developing wearable devices that can monitor muscle fatigue outside of a lab. A huge part of this research for me is that muscle fatigue in a weight lifting sense leads to more muscle growth, but there needs to be a balance as well.

Surface electromyography (sEMG) is commonly used to measure muscle activity. The way it works is by detecting the electrical signals produced when a muscle contracts. Basically, it is shown that as a muscle becomes fatigued, certain characteristics of the EMG signal change. De Luca [1] explained that fatigue causes changes in both the strength, or amplitude, and frequency content of the EMG signal. To be a little more precise, the median frequency of the signal tends to decrease during sustained muscle contractions. De Luca [2] showed that these frequency changes can be used to measure muscle performance and fatigue progression. These findings form the basic idea behind many fatigue detection methods used today.

Cifrek et al. [3] reviewed different ways to analyze muscle fatigue using EMG signals. They described common features such as Root Mean Square (RMS), which measures signal strength, and median frequency, which measures changes in the signal's frequency content. Their review also pointed out that results can be affected by electrode placement, noise, and the type of muscle contraction being performed. This is a huge part of designing the actual wearable device, since wearable sensors are more likely to experience movement and signal interference compared to lab systems.

Researchers have worked on making EMG systems smaller and wearable. Liu et al. [4] developed a wearable EMG patch that can monitor muscle activity and estimate fatigue in real time. This is proof that it is possible to move fatigue monitoring from the lab to a wearable device. The difficult part is that many wearable systems still require calibration or only work well under specific conditions.

Martins et al. [5] reviewed wearable technologies for fatigue monitoring and discussed some of the challenges involved. These include motion artifacts, differences between users, and changes in skin contact during movement. Even through these challenges, simple EMG features like RMS and median frequency can still provide useful information about muscle fatigue when used carefully.

For this research project, I would like to research and develop some kind of wearable that someone could wear at the gym to detect muscle fatigue or muscle contraction. To make it more realistic, I want to focus on the contraction of the bicep. This would focus on the flexion of the muscle and whether a wearable EMG sensor can detect when the muscle is being used. The goal is for the device to record EMG signals and show whether clear contraction trends can be observed during the exercise.

2. Questions and Hypotheses

The goal of this project is to investigate whether a simple wearable EMG system can detect muscle activity in the biceps muscle during a controlled exercise. Previous research has shown that muscle fatigue and contraction cause changes in EMG signals, especially in the strength and frequency of the signal. This project will focus on measuring those changes in the biceps while the muscle is being used repeatedly during an exercise.

Hypothesis 1

Muscle activity in the biceps during repeated contractions will cause measurable changes in EMG signal features over time.

Expected Result

As the biceps is flexed, the EMG signal will show a noticeable change compared to when the arm is relaxed.

Alternative Result

The EMG signal may not show a clear trend if the sensor placement is inconsistent or if the signal contains noise from movement or poor electrode contact.

Interpretation

By placing an EMG sensor on the biceps and recording muscle activity during repeated exercise such as bicep curls or a sustained contraction, the wearable system will allow us to observe how the EMG signal changes as the muscle is used. If consistent patterns appear in the data, it would support the idea that wearable EMG systems can detect bicep muscle contraction outside of a laboratory environment.

Hypothesis 2

A wearable EMG sensor placed on the biceps can consistently measure muscle activity across multiple trials of the same exercise.

Expected Result

When the same exercise is repeated several times, the EMG signal patterns will show similar contraction trends.

Alternative Result

The EMG patterns may vary between trials due to differences in electrode placement, arm movement, or variations in how the exercise is performed.

Hypothesis 3

Basic EMG signal values are enough to detect contraction in the biceps without needing complex machine learning methods.

Expected Result

Simple signal processing will show clear changes in the EMG signal as the biceps muscle contracts.

Alternative Result

The contraction patterns may not be clear enough using only simple features, meaning more advanced analysis methods may be needed.

This project will investigate these hypotheses by building a wearable EMG system that records activity from the biceps muscle during a controlled exercise such as repeated bicep curls or a sustained contraction. The recorded EMG data will then be analyzed over time. By examining how these values change during the exercise, the project will evaluate whether a simple wearable system can detect muscle contraction in the biceps.

3. Device Design

The device for this project used a low-cost EMG sensor connected to an Arduino. The main components were an Arduino Uno, EMG muscle sensor, three electrode pads, jumper wires, a breadboard, resistors, and a laptop for data display.

The EMG sensor was connected to the Arduino using analog pin A0. The signal pin from the EMG sensor was connected to A0 so the Arduino could read the changing EMG values. The power setup used the Arduino 5V pin connected to the red rail of the breadboard and the Arduino ground connected to the blue rail. The EMG +VS pin went to the red rail, the EMG -VS pin went to the blue rail, and the EMG ground was connected to the middle of two resistors.

The electrodes were placed around the bicep area. Two electrodes were placed on the bicep muscle, while the reference electrode was placed near the elbow or forearm. This setup allowed the sensor to detect changes in muscle activity when the bicep was relaxed or flexed.

The wiring setup required some troubleshooting because the sensor was not as simple as a normal three-pin Arduino sensor. However, the breadboard and resistor setup helped create a usable signal for testing.

4. Methods

To test the wearable EMG system, the experiment focused on measuring biceps muscle activity during repeated contractions. First, the electrodes were placed on the subject's biceps muscle. Two electrodes were positioned along the muscle, while a reference electrode was placed on a nearby neutral area such as the elbow or forearm. The EMG sensor was then connected to the Arduino, and the Arduino was connected to the laptop for data viewing.

Once the system was set up, the subject performed a series of controlled bicep contractions. This included simple movements such as flexing the muscle without weight and holding the contraction for a short period of time. The subject was asked to perform the contractions at a steady pace to keep the data consistent.

During the exercise, the EMG sensor detected electrical signals produced by the muscle. These signals were sent to the Arduino as an analog input through pin A0. The values were then displayed using the Arduino Serial Monitor and Serial Plotter.

The experiment was conducted long enough to capture multiple relaxed and flexed states. This allowed changes in muscle activity to be observed over time. After the data was collected, it was reviewed to identify patterns in muscle activation, such as differences between relaxed and flexed states.

To improve accuracy, the electrodes were placed as consistently as possible for each trial, and the subject tried to maintain the same posture and movement speed. Multiple tests were performed to make sure the readings were repeatable.

5. Analysis

The goal of this project was to analyze EMG data collected from the biceps in order to detect when the muscle was being flexed. The EMG signal was collected using the Arduino and viewed through the Serial Monitor and Serial Plotter, where the values could be observed and saved for later use.

The main way the data was analyzed was by looking at how the signal changed over time. When the arm was relaxed, the signal usually stayed around the upper 800s. When the bicep was flexed, the signal dropped, sometimes as low as about 650. By looking at the signal over time, it was possible to see clear differences between when the muscle was relaxed and when it was active.

To make this easier to detect, a simple threshold was used. Based on the readings, a value of about 830 was chosen as the cutoff. If the EMG value dropped below 830, it was counted as a flexed or contraction state. If the value stayed above 830, it was counted as relaxed. This gave a simple way to classify when the muscle was being used.

The results from this analysis were compared to the original hypothesis. Since the signal consistently changed during flexion and returned closer to the relaxed range afterward, this supported the idea that the system could detect bicep muscle contraction.

There were some limitations to this approach. The EMG signal could be affected by noise, movement, or how well the electrodes were placed on the skin. These factors made the signal less consistent at times. Also, using a simple threshold may not capture smaller or more subtle muscle activity.

The code used to collect and display the EMG data will be included in the project repository. This code handled reading the signal from the sensor and outputting values that could be used for analysis.

6. Discussion of Findings

The main finding from this project was that the EMG sensor was able to show a noticeable signal change when the bicep was flexed compared to when it was relaxed. In this setup, the relaxed signal was usually higher, around the upper 800s, while flexing the bicep caused the

value to drop, sometimes as low as about 650. Because of this, the analysis used a simple threshold method: if the EMG value dropped below about 830, it was counted as a flexed/contraction state, and if the value stayed above 830, it was counted as a relaxed state.

This means the sensor did not need to perfectly measure muscle strength, but it still showed a clear enough difference to detect when the bicep was being used. The biggest issues were noise, wiring, and electrode placement, which made the readings change at times. The budget friendly EMG sensor made the project less accurate than more polished devices like the MyoWare Muscle Sensor, Grove EMG Detector, BITalino, or OpenBCI, which are designed to collect cleaner EMG or biosignal data [6]-[9]. Even with those limitations, the project worked as a basic proof of concept because the Arduino and EMG sensor were able to separate relaxed and flexed muscle states using the recorded signal values.

7. Future Implementations

There are several ways this project could be improved in the future.

First, the device could be made into an actual wearable system. Right now, the setup uses wires, a breadboard, and a laptop, which works for testing but is not very practical for real use. A future version could place the Arduino, sensor, and wiring into a small case or strap.

Second, different types of curls could be tested. A normal bicep curl and hammer curl may produce different EMG readings because they use the biceps and nearby muscles differently.

Third, the system could be tested on other muscle groups. For example, it could be used on the forearm, calf, or shoulder to see if the same basic method works.

Fourth, an LED indicator could be added. The LED could turn on when the EMG value drops below the flex threshold, giving a simple visual signal when the muscle is being used.

Fifth, a simple app or display could be added. Instead of only using the Serial Monitor, the system could show “relaxed” or “flexed” in real time.

Finally, a better EMG sensor could be used. A higher quality sensor would likely produce cleaner data and reduce the problems caused by noise and electrode placement.

8. Conclusion

Overall, this project showed that a simple Arduino and EMG sensor setup can detect bicep muscle contraction. The system was able to read muscle activity, display signal changes, and show differences between relaxed and flexed states.

The project was not perfect, and there were several issues with noise, electrode placement, and wiring. However, the final results showed that the sensor values changed enough during flexion to separate relaxed and flexed muscle states using a basic threshold. With better hardware, cleaner wiring, and more consistent electrode placement, this system could become more reliable and useful for exercise tracking, rehabilitation, or muscle-controlled device projects.

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